Twinlab: a patient Digital Twin

A two-hour glucose forecast and spike-risk prototype

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Team members

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Problem and healthcare use case

Condition

Type 2 Diabetes; focus on a new elevated-glucose endpoint.

Use case

A doctor inspects combined patient history and sensor data, reviews an early risk flag, and checks the explanation.

Scope

A focused patient-state replica, not a whole-body physiology or treatment model.

A precise prediction target

Forecast horizon

Exactly 120 minutes after each forecast is issued.

New event

Current glucose is at or below 180 mg/dL; observed glucose two hours later is above 180 mg/dL.

Separate outputs

One model predicts the value. A dedicated classifier estimates event risk. A risk alert can occur below the point-forecast threshold.

Two separate data streams

Static EHR

Age, sex, diagnosis duration, HbA1c, BMI and a hypothetical genetic marker.

Dynamic sensors

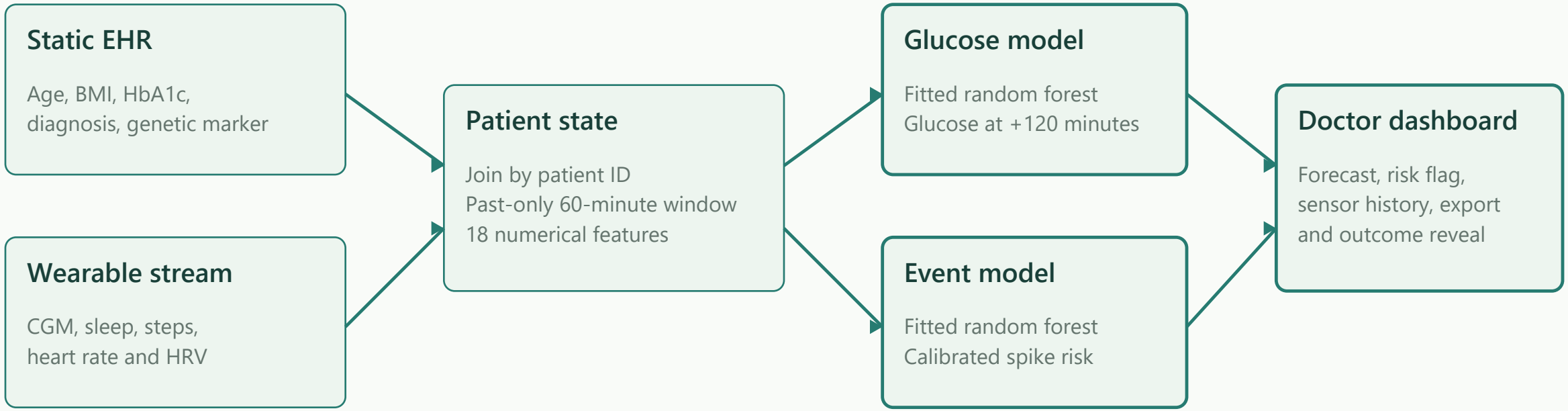
Five-minute glucose, sleep duration, steps, heart rate and HRV. Values change as the patient timeline advances.

Data sandbox

90 fully synthetic people, 8 days each. No real patient records or identifiable health data.

System architecture

Both streams update the virtual patient; future readings are reserved for evaluation



Future observations are held back from inference. They are revealed only to check the saved prediction.

An updated virtual patient

Align and summarize

Join by patient ID; compute past glucose lags, mean, spread, slope and recent activity.

Predict at each simulated event

Refresh the 18-feature patient state and run exported fitted models.

Inspect and compare

History, risk, uncertainty band and feature sensitivity appear on the clinician dashboard.

Seven benchmark methods

Glucose forecasting

Persistence, 30-minute linear trend, ridge regression, random forest and gradient boosting.

Spike detection

Logistic regression and random forest classifiers; cutoff selected on validation F2 to favour recall.

Technical stack

Python 3.13, NumPy, scikit-learn; browser JavaScript, HTML, CSS and SVG. Model artifacts are JSON.

Patient-separated evaluation

Patient cohort	Count	Purpose
Training	54	Fit the models
Validation	12	Choose models / threshold
Calibration	12	Risk mapping / prediction band
Test	12	4,416 final forecasts

No patient appears in more than one cohort. Only current and past observations form the features.

Glucose forecast results

4,416 predictions on 12 unseen synthetic patients; lower error is better

Method	Average error	RMSE
Ridge fusion	14.4 mg/dL	18.7 mg/dL
Random forest fusion	10.0 mg/dL	14.0 mg/dL
Gradient boosting fusion	10.3 mg/dL	14.2 mg/dL
Persistence (last glucose)	25.6 mg/dL	32.9 mg/dL
Linear trend (30-minute slope)	45.9 mg/dL	61.8 mg/dL

Alert performance and errors

Dedicated event classifier; synthetic performance only

Event classifier	Spikes caught	Alerts correct
Logistic spike classifier	78.0%	32.1%
Random forest spike classifier	71.3%	35.3%

Deployed classifier: 211 correct alerts, 386 false alarms, 85 missed events and 3444 correct non-alerts.

The model and cutoff were selected on validation patients. Scores count forecast windows, not independent clinical episodes.

Live demonstration

1. Predict

Open the dashboard; choose a patient, time and sensor quality.

2. Reveal

Choose a measured example and reveal the actual +2-hour outcome. The saved prediction stays fixed.

3. Verify

Inspect a correct alert, false alarm and miss. Export results and compare the complete benchmark report.

Robustness and reproducibility

Additional benchmarks

Patient-group cross-validation, chronological holdout, one-stream ablations, sensor gaps, noise and subgroup diagnostics.

Uncertainty

Separate calibration patients; 89.7% empirical test interval coverage. Overlapping windows limit formal coverage claims.

Independent checks

All 4,416 Python/browser outputs agree; 207,360 sensor records and all future labels pass integrity checks.

Limits and research next steps

Observed limitation

Only 35.3% of alerts are correct. False alarms remain substantial.

What is not established

Clinical effectiveness, real genetic associations, treatment effects and real-world patient safety.

Next research

Use appropriately permitted longitudinal data; pre-register targets; run prospective, patient-separated validation.

Repository and submission

Public repository

github.com/Paras-so/Digital-Twin-Challenge-2026

Included materials

Source, fitted models, synthetic input streams, test predictions, MIT license, architecture and presentation PDFs.

Video status

PENDING: record and link a minimum 20-minute explanation and demonstration. A recording guide is supplied.